

AE-349: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH, shorter irradiation

Date: 2024-09-16

Tags: Dihydroxo Radiation PNP
P(tBu)N(py)P(tBu) [Ru(CO)(OH)₂(PNP)]
NMR AE 1H 31P

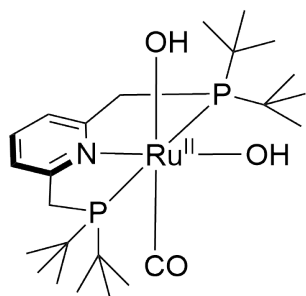
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Status: Done

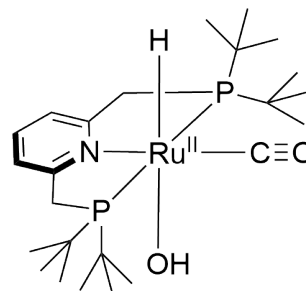
Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



365 nm (0.3 W)
white (0.2 W)
H₂O



Chemical Formula: C₂₄H₄₅NO₃P₂Ru

Molecular Weight: 558,64552

[Ru(CO)(OH)₂(PNP^{tBu})]

Chemical Formula: C₂₄H₄₅NO₂P₂Ru

Molecular Weight: 542,64652

[Ru(CO)H(OH)(PNP^{tBu})]

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiment	Experiment - AE-166: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})], 92 % pure, 365 nm and white, AE-246: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-238, 97 % pure) 365 nm and white AE-348: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, water, longer irradiation

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH) ₂ (PNP{tBu})]	AE-310: Synthesis of [Ru(CO)(OH) ₂ (PNP ^{tBu})] using Ag ₂ O, 5 h (AE-310-1)	5.73	/	/	3.20	558.65	/
Milli-Q water	KRA-104: Degassing of Milli-Q water	/	/	/	/	/	1.00

KOH (1M)	AE-243: 1 M Stocksolution of KOH in water	/	/	/	/	/	0.05
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Irradiation Parameters

Power measured using [\[Power Meter\] 843-R-USB + 919P-020-12](#) unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9.5
Thermostat temperature [°C]	25

Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#). All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
16.09		A 10 mL flask was prepared	
	9:25	To the flask [Ru] was added under air	
		The flask was set under argon	
	9:41	To the flask water was added	
	9:43	To the flask KOH (1M) was added using a 100 μ L Hamilton syringe	A 50 mM KOH solution was obtained
	9:45 - 9:55	The obtained mixture was stirred at 200 rpm.	
	9:55	From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D2O/H3PO4 capillary. (AE-349-1) During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 μ m)	AE-349-1.jpg
	14:15	The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to [Protocol] Irradiation setup	power.jpg
	14:20	The NMR-tube was placed in the setup accroding to [Protocol] Irradiation setup	

	14:23 - 15:23	The J.Young NMR tube was irradiated for 1:00 h	first irradi.jpg
	15:29	An NMR was measured (AE-349-1-1)	clear orange, liquid AE-349-1-1.jpg
17.09		The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	
	11:57 - 13:57	The J.Young NMR tube was irradiated for 2:00 h	second irradi.jpg
	14:35	An NMR was measured (AE-349-1-2)	clear orange, liquid, small amount of solids AE-349-1-2.jpg
18.09		The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	
	11:37 - 13:37	The J.Young NMR tube was irradiated for 2:00 h	third irradi.jpg
	14:18	An NMR was measured (AE-349-1-3)	clear orange, liquid, small amount of solids AE-349-1-3.jpg
19.09		The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	
	10:39 - 12:39	The J.Young NMR tube was irradiated for 2:00 h	4th irradi.jpg
	13:03	An NMR was measured (AE-349-1-4)	clear orange, liquid, small amount of solids AE-349-1-4.jpg
09.10	14:30	An NMR was measured (AE-349-2)	clear orange liquid, significant amount of solids at the bottom AE-349-3.jpg
11.10		An NMR was measured (AE-349-3)	clear orange liquid, significant amount of solids at the bottom
12.11		An NMR was measured (AE-349-4)	clear orange liquid, significant amount of solids at the bottom AE-349-4.jpg
14.11		The irradiation setup from Two-photon - AE-378: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 5 mM Na2HPO4 was used without any modification	
		The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	
	16:46 - 21:46	The J.Young NMR tube was irradiated for 5:00 h	irradi setup.jpg
		An NMR was measured (AE-349-5)	clear dark orange-brown liquid, significant amount of solids at the bottom AE-349-5.jpg

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
16.09	/	AE-349-1	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-349-1.zip	AE-349-1.zip_AE-349-1_103.nmrium	/	Reactant signal is gone after 5 h of irradiation. Hydride species at 90 ppm is formed
16.09	/	AE-349-1-1	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-349-1-1.zip	AE-349-1.zip_AE-349-1_103.nmrium	/	increasingly until the reactant is consumed Side products: 55 ppm: is in starting material and does not change
17.09	/	AE-349-1-2	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-349-1-2.zip	AE-349-1.zip_AE-349-1_103.nmrium	/	significantly 57 ppm: Main side product increases till all reactant is consumed
18.09	/	AE-349-1-3	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-349-1-3.zip	AE-349-1.zip_AE-349-1_103.nmrium	/	58.5 ppm: small amount, forms under irradiation 62 ppm: very small only visible after 7 h
18.09	/	AE-349-1-4	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-349-1-4.zip	AE-349-1.zip_AE-349-1_103.nmrium	/	65 ppm: small only visible after 1 h 74.5 ppm: small only visible after 3 h
09.10	/	AE-349-2	1H, 1Hzgespg, 31P{1H}quant, 1H/31P-HMBC	IAAC 400 MHz I	H2O	AE-349-2.zip	/	AE-349-1 and AE-349-1-4	New hydride signal in 1H at -18.4 ppm. Reactant observed in 31P New signal in 31P at 81.8 ppm --> maybe corresponds to new hydride signal HMBC: only measured from 8 to -0.5 ppm in 1H --> no correlation of hydride signals. Coupling of P with methyl and methylene groups observable for various species. Further evaluation needed.

11.10	/	AE-349-3 (in ZIP and NMRIUM file labeled as AE-348-3)	1H, 1Hzgespg, 31P{1H}quant, 1H/31P-HMBC	IAAC 400 MHz I	H2O	AE-349-3.zip	2024-10-14_8-20-58.zip_AE-349-3_101.nmrium	https://doi.org/10.1039/D1E01053K , AE-349-1 and AE-349-1-4	Detailed evalutaion of HMBC: Methyl area: for each main species (90.5, 81.8, 66.9 and 56.7) signals are observed. Two signals are observed at 90 and 82 ppm --> fits expected, since hydride species and not symetric. Other two have one signal --> symetric. Also signals observed for further minor side products. Hydride signals: 90.6 and -6.3 fit to known species. New signal at 81.8 and -18.4 clearly seen (fits to trans species in compariosn to PNN. Methylene area: stronger coupling observed for signals at 91 and 58 ppm. Each 31P signal has one signal in 1H. Shifts are quite similar: 90.6 3.48, 81.8 3.40, 66.9 2 signals observed, but one may be from minor impurity: 3.53 and 2.95 56.8 3.32. No signals in aromatic area observed.
12.11	22:16	AE-349-4	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-349-4.zip	AE-349-5_10.nmrium	AE-349-3	Only minor changes. Slightly more trans isomer and nearly no major side product observed.
15.11	12:57	AE-349-5	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-349-5.zip	AE-349-5_10.nmrium	AE-349-4 and AE-349-1	Generally as expected. Nearly no trans species observed. No reactant observed, but 2 side products very close to it. Large signal of main side porduct observed.

Results

Sample	Yield (hydride species) [%]	Yield (dihydroxo) [%]	Data
AE-349-1-1	4	69	AE-349_Yield_Determination.xlsx
AE-349-1-2	18	23	AE-349_Yield_Determination.xlsx

AE-349-1-3	26	0	AE-349_Yield_Determination.xlsx
AE-349-1-4	27	0	AE-349_Yield_Determination.xlsx

Sample	Yield (hydride species, cis) [%]	Yield (hydride species, trans) [%]	Yield (dihydroxo) [%]	Yield (main side prod) [%]	Remaining stuff [%]	Data
AE-349-1-4 (after 7 h)	26	0	0	46	28	AE-349-3_Yield_Determination.xlsx
AE-349-3 (after 23 d)	18	16	17	14	36	AE-349-3_Yield_Determination.xlsx
AE-349-5 (after 2nd irradiation)	15	1	0	34	49	AE-349-5_Yield_Determination.xlsx

Irrad over time: full conversion after 5 h, no significant change by additional 2 h of irradiation.

After 23 d at rt in dark: isomerisations happen and increase of hydride yield: cis hydride decreases slightly to 18 %, trans formed with 16 % increasing total hydride yield to 34 %. Side product decreases very significantly to 14 %. Dihydroxo is formed again with 17 %. Not clear what is formed from what.

2nd irradiation: Seems like no new hydride is formed, since the hydride yield decreases significantly. In comparison to AE-349-4 a bit more cis hydride is formed (could be due to irradiation), but nearly all trans hydride is gone. Maybe the trans hydride decomposes under irradiation

Linked experiments

- AEI-022: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNN})]$, NO₂-route, 365 nm and white LED
- AEI-026: Capillary production
- AEI-064: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag₂O
- AEI-067: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, crude, 365 nm and white
- AEI-071: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purer, 365 nm and white
- AE-105: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag₂O
- AE-110: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white
- AE-119: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 52 mM KOH, 365 nm and white
- AE-140: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white, longer duration I
- AE-145: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag₂O
- AE-158: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white
- KRA-092: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$, 365 nm and white
- KRA-104: Degassing of Milli-Q water
- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag₂O, 5 h
- AE-243: 1 M Stocksolution of KOH in water
- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white
- AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag₂O, 5 h
- Two-photon - AE-348: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, water, longer irradiation
- Two-photon - AE-378: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 5 mM Na₂HPO₄

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Light Source - [UHP LED 365 nm-1](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Schlenk Technique](#)

Protocol - [Irradiation setup](#)

Attached files

AE-349-5_Yield_Determination.xlsx

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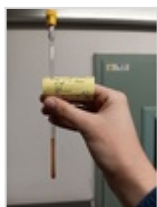
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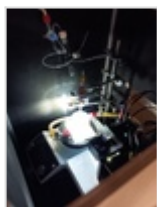
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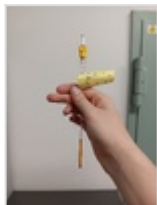
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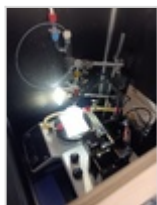
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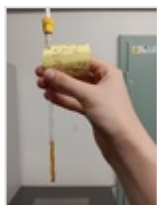
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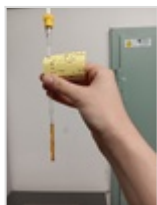
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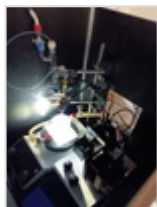
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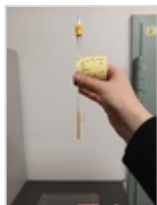
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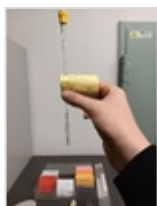
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Comment

On 2024-11-19 14:20:49 Jacob Schneidewind wrote:

Notes:

* adding analysis for -4 in overview table



Unique eLabID: 20240916-38efa916818935b9d962cc92a0f27f811761f1bb

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1356>